Task 7

Dashboards and Reports for Prediction Analysis

Predictive Analytics & AI Model Deployment - Customer Churn Prediction

Objective. Create dashboards and reports that explain model predictions to business users.

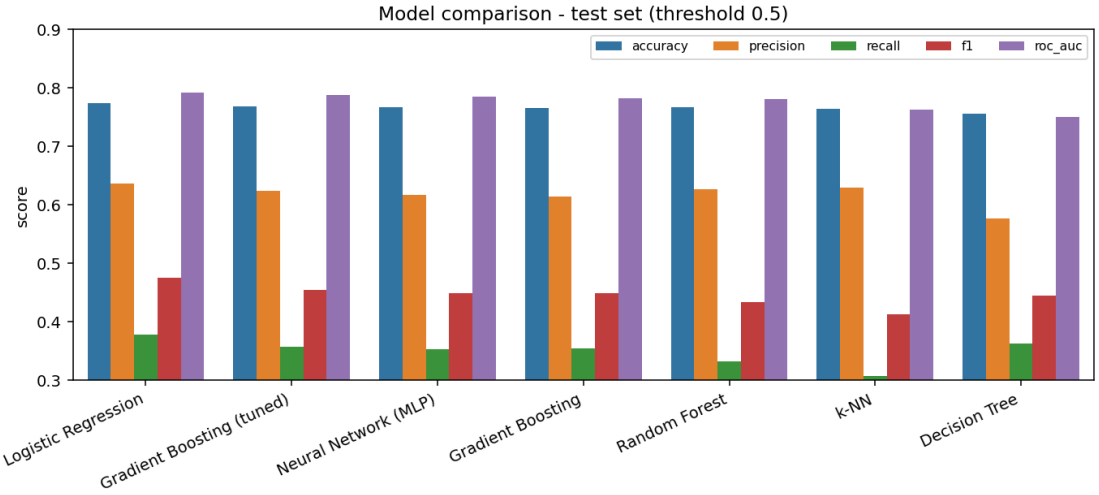
1. What was built

A single self-contained HTML dashboard (`reports/dashboard.html`, 0.9 MB, charts embedded as images, no internet needed) plus the `reports/test_predictions.csv` file behind it. It is generated by `src/build_dashboard.py` and served by the API at `/dashboard`.

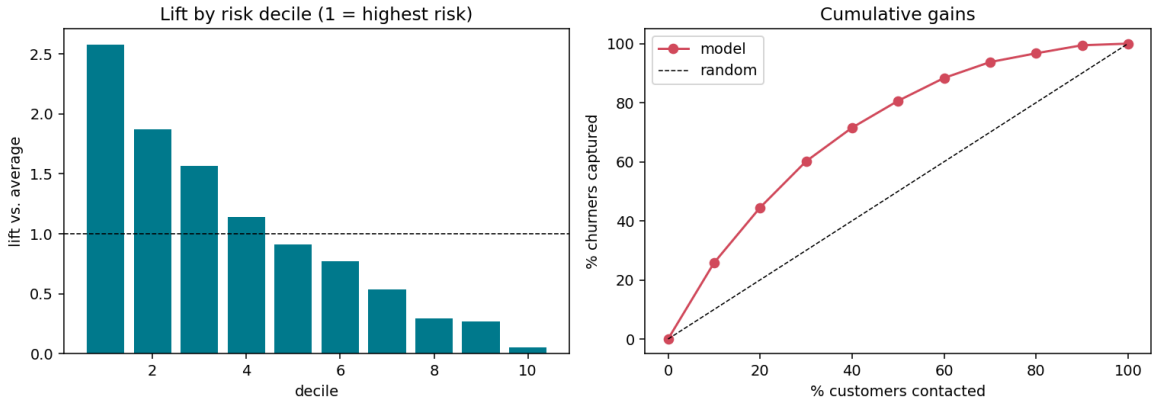
Dashboard sections

- **KPI cards**: customers analysed, overall churn rate, best model, test ROC-AUC, recall at the chosen threshold, number of high-risk customers.
- **Data and drivers**: churn by contract / tenure / payment; correlation with churn.
- **Model comparison**: metrics table, grouped bar chart and ROC curves for all models.
- **Prediction analysis**: confusion matrix + calibration, permutation importance, lift and gains, predicted-vs-actual by risk segment.
- **Risk segments** table and an illustrative **retention-campaign ROI** table.
- **Top-15 highest-risk customers** in the test set (a ready-made call list).
- **Neural-network training curves**.

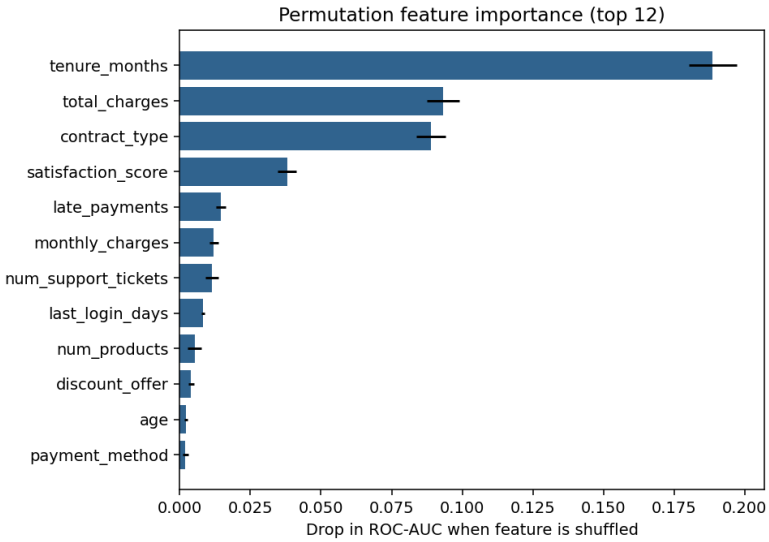
2. Key charts reproduced from the dashboard



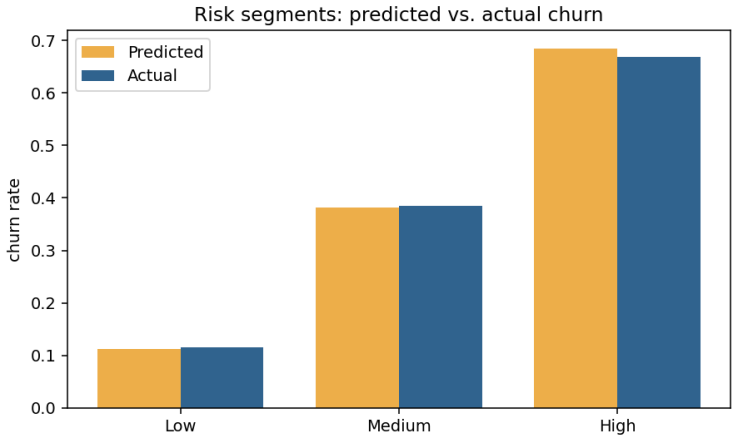
Model comparison chart



Lift and cumulative gains



Feature importance



Risk segments

3. Reading the dashboard - prediction analysis summary

- The selected Logistic Regression model ranks customers well: ROC-AUC 0.791, top-decile lift 2.58x, and the top 20% of customers hold 44.5% of churners.
- Tenure, contract type and total charges are the main drivers; satisfaction score and payment behaviour add smaller effects.

- Predicted and actual churn rates match in each risk segment, so the segment labels can be trusted operationally.
- Recommended use: contact High-risk customers first; review Medium risk when capacity allows.

4. Output files

- `reports/dashboard.html` - open in any browser.
- `reports/figures/*.png` - individual charts.
- `reports/test_predictions.csv` - probability, decision and segment per test customer.

Appendix - Source code

src/build_dashboard.py

```
"""TASK 7 - Build a self-contained HTML dashboard + prediction-analysis report."""
import base64
import json
import pandas as pd
from .config import FIGS, MODELS, REPORTS, RESULTS


def img(name, alt=""):
    b = base64.b64encode((FIGS / name).read_bytes()).decode()
    return f'<img alt="{alt}" src="{b}">'


def table(df, cls=""):
    return df.to_html(index=False, border=0, classes=cls, float_format=lambda v: f"{v:.3f}")


def main():
    meta = json.loads((MODELS / "metadata.json").read_text())
    t1 = json.loads((RESULTS / "task1_collection.json").read_text())
    t2 = json.loads((RESULTS / "task2_preprocess.json").read_text())
    t4 = json.loads((RESULTS / "task4_comparison.json").read_text())
    t5 = json.loads((RESULTS / "task5_analytics.json").read_text())
    pred = pd.read_csv(REPORTS / "test_predictions.csv")
    m = meta["test_metrics_at_threshold"]
    high = int((pred["risk_segment"] == "High").sum())

    cmp = pd.DataFrame(t4["table"])[["name", "accuracy", "precision", "recall", "f1", "roc_auc", "pr_auc"]]
    cmp.columns = ["Model", "Accuracy", "Precision", "Recall", "F1", "ROC-AUC", "PR-AUC"]
    seg = pd.DataFrame(t5["segments"])[["risk_segment", "customers", "share", "avg_pred", "actual_rate"]]
    seg.columns = ["Segment", "Customers", "Share", "Avg predicted", "Actual churn"]
    roi = pd.DataFrame(t5["roi"]); roi.columns = ["Top % contacted", "Contacted", "Churners reached", "Expected saved", "Net
        value ($)"]
    top = pred.sort_values("churn_probability", ascending=False).head(15)[["customer_id", "churn_probability", "risk_segment",
        "actual_churn"]]
    top.columns = ["Customer", "Churn probability", "Segment", "Actually churned"]

    kpis = [("Customers analysed", f"{t2['clean_rows']:,}"), ("Overall churn rate", f"{t2['churn_rate_train']*100:.1f}%"),
            ("Best model", meta["best_model_name"]), ("ROC-AUC (test)", f"{m['roc_auc']:.3f}"),
            ("Recall @ thr %.2f" % meta["threshold"], f"{m['recall']*100:.0f}%"), ("High-risk customers (test)", f"{high:,}")]
    cards = "".join(f'<div class="kpi"><div class="v">{v}</div><div class="l">{k}</div></div>' for k, v in kpis)
    html = f"""<!doctype html><html><head><meta charset="utf-8"><meta name="viewport" content="width=device-width,initial-
        scale=1">
<title>Churn Prediction Dashboard</title><style>
body{{font-family:system-ui,Segoe UI,Arial,sans-serif;background:#f4f6f8;margin:0;color:#1d2733}}
header{{background:#1f3a5f;color:#fff;padding:20px 28px}}h1{{margin:0;font-size:22px}}header p{{margin:4px 0 0;opacity:.8;font-
        size:13px}}
main{{max-width:1150px;margin:18px auto;padding:0 16px}}h2{{font-size:17px;margin:26px 0 8px;color:#1f3a5f}}
.kpis{{display:grid;grid-template-columns:repeat(auto-fit,minmax(170px,1fr));gap:12px}}
.kpi{{background:#fff;border-radius:10px;padding:16px;box-shadow:0 1px 4px rgba(0,0,0,.08)}}.kpi .v{{font-size:22px;font-
        weight:700;color:#1f3a5f}}.kpi .l{{font-size:12px;color:#5c6b7a;margin-top:4px}}
.grid{{display:grid;grid-template-columns:repeat(auto-fit,minmax(480px,1fr));gap:14px}}
.card{{background:#fff;border-radius:10px;padding:14px;box-shadow:0 1px 4px rgba(0,0,0,.08);overflow-x:auto}}.card
        img{{width:100%;height:auto}}
table{{border-collapse:collapse;width:100%;font-size:13px}}th{{background:#e9eef4;text-align:left}}th,td{{padding:6px
        9px;border-bottom:1px solid #e3e8ee}}
.note{{font-size:12px;color:#5c6b7a;margin-top:8px}}
</style></head><body><header><h1>Predictive Analytics &amp; AI Model Deployment - Churn Dashboard</h1>
<p>Best model: {meta['best_model_name']} &middot; threshold {meta['threshold']} &middot; generated {meta['trained_at'][:10]}
        &middot; data: synthetic 30,000-customer subscription dataset</p></header><main>
<div class="kpis">{cards}</div>
<h2>1. Data &amp; drivers</h2><div class="grid"><div class="card">{img('t2_churn_by_segment.png')}</div><div
        class="card">{img('t2_target_corr.png')}</div></div>
<h2>2. Model comparison</h2><div class="card">{table(cmp)}<p class="note">Held-out test set of {t4['test_rows']:,} customers;
        metrics at threshold 0.5. Majority-class accuracy baseline = {t4['majority_baseline_accuracy']:.3f}.</p></div>
<div class="grid" style="margin-top:14px"><div class="card">{img('t4_metric_bars.png')}</div><div
        class="card">{img('t4_roc.png')}</div></div>
<h2>3. Prediction analysis</h2><div class="grid"><div class="card">{img('t4_confusion_calibration.png')}</div><div
        class="card">{img('t5_importance.png')}</div>
<div class="card">{img('t5_lift_gains.png')}</div><div class="card">{img('t5_segments.png')}</div></div>
<h2>4. Risk segments</h2><div class="card">{table(seg)}</div>
<h2>5. Retention campaign simulation (illustrative)</h2><div class="card">{table(roi)}
<p class="note">Assumptions: ${t5['assumptions']['cost_per_offer']:.0f} per offer,
        ${t5['assumptions']['value_of_saved_customer']:.0f} value per saved customer, {t5['assumptions']['save_rate']*100:.0f}%
        of contacted churners retained. These are placeholders - replace with real business numbers.</p></div>
<h2>6. Highest-risk customers in the test set</h2><div class="card">{table(top)}</div>
<h2>7. Neural network training</h2><div class="card">{img('t5_mlp_training.png')}</div>
</main></body></html>"""
    (REPORTS / "dashboard.html").write_text(html, encoding="utf-8")
    print("dashboard.html", round(len(html) / 1e6, 2), "MB")


if __name__ == "__main__":
    main()
```